WHAT NORMALIZATION'S NON-ASSOCIATIVITY IS — AND IS *NOT*: THREE OBSTRUCTION THEORIES THAT MISS IT FOR ONE REASON, AND THE IDEMPOTENT NORMALIZER — CLASSICALLY, A TRANSVERSAL — THAT RECEIVES IT

MACBETH

Abstract. The normalization of substochastic kernels is non-associative: Di Lavore, Román and Székely (arXiv:2510.01131) show it is a *sesquilaw*, not a distributive law, and that the induced composition of normalized kernels forms a genuine *magmoid* **Norm** — a category minus associativity. We ask which categorical invariant measures that defect. The question is a probabilistic, finite-dimensional face of a central concern of the program: orchestration composition is governed by the Zappa–Szép *weld* of directed containers (our tool for *when* compositions exist), which *is* associative once its Zappa–Szép data make it well defined; **Norm** is an exact-factorization cousin of the weld and a fully computable witness of the non-associativity that *normalization* introduces on top. We answer in two movements. *Movement I* refutes three *named* candidate homes — a 2-cocycle, a Karoubi/Cauchy splitting obstruction, and a Schoen/Tholen lax distributive law — and shows they fail for *one* reason, read three ways: the renormalization idempotent k is a *unary* 1-cell that never sees a triple, whereas non-associativity is irreducibly *ternary*. *Movement II* identifies the invariant: it is received by an idempotent *normalizer* K obeying a left- but not right-absorption law. Writing $f \# g := K(f \cdot g)$, associativity of $\#$ is equivalent to K being two-sided absorbing; and the decisive, directly checkable consequence is a *closure criterion — if* **Norm** *is handed then* $\operatorname{Im} K$ *is not closed under composition.* This is the forward direction of the classical transversal/right-loop theorem, and it refutes a fourth named map (the near-semiring's one-sided absorbing zero) at a stroke: a principal normalizer has closed image, hence is associative. We are explicit about provenance: the normalizer package is the classical transversal construction of right loops (Baer; Lal; Kiechle), published already in oidified form (Grabowski); $\mathbf{Norm} = D \cdot K$ is an *exact factorization* — one honest rung more general than a Zappa–Szép product, in which only the subgroup factor, not the transversal, is closed. The honest novelty is narrow: a base *without* invertibility, and the substochastic-kernel instance. A machine-checked finite model certifies the construction axiom-free.

1. Setup: the normalization defect, and the weld that motivates it

The object of this note is a concrete non-associative magmoid; the weld is its motivation. We keep the two cleanly apart, because conflating them was the first thing a referee (rightly) struck.

The motivation. Throughout, **Poly** is the category of polynomial functors on **Set** with the composition monoidal product $\triangleleft$; a comonoid for $\triangleleft$ is a *polynomial comonad*, equivalently a small category. For a set S we write $\Delta S = S\mathsf{y}^S$ (notation y for the representable) for the *store* comonad on S, the directed container of the full state space on S. All composites are written in diagrammatic order: $f\,;g$ (equivalently $f \cdot g$) means "f then g".

Definition 1.1 (The weld)**.** Given two directed containers presented by store comonads ΔS (the *worker* state) and ΔT (the *harness* state), their *weld* is the orchestration composite $B = \Delta S \triangleleft \Delta T$,

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the Zappa–Szép product interleaving worker and harness transitions. The weld is the program's canonical answer to "when does a composite of two stateful directed containers exist?": the Zappa–Szép datum is exactly the mutual-action data that makes the interleaving well defined.

Two facts about the weld frame — and bound — what follows.

Remark 1.2 (The weld, once well defined, is *associative*; keep it separate). When its Zappa–Szép data make it well defined, the weld is a genuine comodule, and in that regime it is associative. Concretely it is the *regular* $(\Delta T, \Delta S)$-bicomodule, which under Spivak–Garner–Fairbanks ("Functorial Aggregation", arXiv:2111.10968, Prop. 3.9(5) + Thm. 4.28) represents the parametric right adjoint $\bar{F}_B(V) = (T \times V)^S = T^S \cdot V^S$ (registry `weld-is-regular-bicomodule`, **peer-reviewed**, with a Lean-verified leaf for the five coaction axioms). The cohomological class $[\omega] \in H^2$ attached to a well-defined weld (registry `g-obstruction-cohomology`, in `pairwise-zs.json`) is a *splitting* obstruction of that *already-associative* weld — it measures whether the extension splits, not whether the composite associates. Neither the bicomodule picture nor $[\omega]$ is what we chase here. The present defect is the *departure* from the associative regime that normalization introduces, and we study it in its own clean habitat.

Remark 1.3 (How **Norm** and the weld are related — and how they are not). We do *not* claim **Norm** (below) computes the weld's associativity defect, nor that there is a structure map weld-triples $\to$ **Norm**-triples; we make no such map and need none. The link is of a different, honest kind: both are instances of *Zappa–Szép / exact-factorization* composition. The weld is the Zappa–Szép product of two directed containers; and **Norm**, we will see (§9), is the exact factorization $D \cdot K$ of normalized kernels read probabilistically. **Norm** is thus a finite-dimensional, fully computable, machine-checkable witness of *the kind* of non-associativity that orchestration composition in the program must contend with. That is why it is worth measuring; it is not a stand-in for the weld.

The object. Di Lavore, Román and Székely ("Normalized Probabilistic Semantics is Not Associative", arXiv:2510.01131) show that *normalization* of substochastic kernels is a *sesquilaw*, not a distributive law, and that the induced Kleisli-style composition of normalized kernels is a genuine magmoid.

Definition 1.4 (Sesquilaw and the renormalization idempotent, after arXiv:2510.01131)**.** A *sesquilaw* (S, T, m, n) consists of a distributive law $m\colon ST \Rightarrow TS$ together with an *almost-distributive* law $n\colon TS \Rightarrow ST$ satisfying all four distributive-law axioms *except* multiplicativity, which holds only up to the natural transformation

$$k \;=\; n\,;m \;:\; TS \Rightarrow TS \qquad \text{(the "renormalization" of Lemma 3.5).}$$

For normalization one takes $T = D$ (distribution) and $S = M$ (maybe), so $TS = DM$ and $ST = MD$; then $n = (-)^\circ\colon DM \Rightarrow MD$ is normalization and $m = (-)^\bullet\colon MD \Rightarrow DM$ is the subdistribution inclusion. The induced magmoid **Norm** of normalized kernels $X \to MDY$ is a category minus associativity; Proposition 4.9 of loc. cit. isolates the triples that do associate. *The question of this note is: which categorical invariant is the non-associativity of* **Norm***?*

2. The method: name the map, then decide

We adopt one discipline, learned the hard way across a month of cross-domain bridge-hopes that died on inspection.

Remark 2.1 (Naming the map buys decidability, not survival). A candidate obstruction home ranked highly because its *vocabulary* matches ("cocycle", "splitting", "Yang–Baxter") has earned nothing until a *structure map* is exhibited between the *specific* objects at hand — not the words. "This sounds like it should connect" is a prior of roughly zero. The payoff of naming a candidate map is not that it survives; it is that the question becomes *decidable*. Our record on resonance-ranked leads is $0/n$; our record on *named*-map leads is: all decided, crisply, one way or the other.

The note has two movements. In the first (§3) we refute three candidate homes and show the refutations are not three coincidences but one fact read three ways. In the second (§§4–9) we exhibit the structure that *does* receive the invariant, derive the decisive closure criterion, and watch a fourth candidate fall out of it. Each verdict is one of three epistemic kinds, kept strictly separate:

- **proved** — a registered result of the program (the refutations and the positive construction; several carry Lean-verified leaves; the Movement II algebra is peer-reviewed);
- **computed** — a corroborating but non-formalized numerical fact (so flagged in place);
- **open** — an explicitly unresolved question (so flagged).

Provenance of the *surviving* structure is handled separately and honestly in §9: it is classical, and we say so.

Movement I. One reason, three readings

3. One reason, three readings: the renormalization idempotent is the wrong arity

Three named maps have been proposed as the home of **Norm**'s non-associativity. Each is refuted below. But they do not fail independently; they fail for a single reason, which we state first so the three readings can be seen as what they are.

Remark 3.1 (The one reason). Every one of the three candidates is built from the renormalization idempotent $k = n\,;m$ — a *single* 1-cell $DM \Rightarrow DM$ in a functor category. A 1-cell is *unary* data: it takes one input. Non-associativity, by contrast, is irreducibly *ternary*: it is the assertion that the associator

$$Z_K(f,g,h) \;:=\; \big((f\,\#\,g)\,\#\,h,\; f\,\#\,(g\,\#\,h)\big)$$

has unequal components for some triple (f,g,h). No feature of a single unary cell — whether it is a coboundary, whether it splits, which way an inequality on it points — is a function of a triple; so none of them can *be* the associator. The three refutations below are three ways of saying this: cocycle (wrong *arity*), Karoubi (a unary operation that preserves associativity), lax-DL (a fixed-direction order on one cell, blind to a ternary incomparability). The invariant lives one level up, at the argument slots of the binary #, where Movement II places it.

3.1. **Reading 1: not a 2-cocycle — wrong arity.**

Claim 3.2 (registry: refuted; cf. `h2-weld-class-is-splitting-not-associativity`, `k-as-2-cocycle`). The renormalization idempotent $k = n\,;m$ is not a 2-cocycle. The refutation is one of *type*: k has the wrong arity.

Why the type is wrong. A 2-cocycle is a *binary* gadget whose cocycle identity is an equation among the *three* ways of associating a triple; it is, by construction, a function that sees triples. The renormalization k is a single 1-cell $DM \Rightarrow DM$ — unary data. There is no coercion of a fixed unary cell into a binary-cum-ternary cocycle; the arities do not match, so the question "is k a 2-cocycle?" is not false but *ill-typed*. (This corrects an earlier, stronger and incorrect claim of ours — "non-associativity cannot be carried as a cohomology class." It can; see Remark 3.3.) □

Remark 3.3 (Non-associativity *can* be cohomological — but in degree 3). The octonions are the twisted group algebra $k_F[\mathbb{Z}_2^3]$ whose associator $\varphi = \partial F$ is a genuine 3-*cocycle* (Albuquerque–Majid, "Quasialgebra Structure of the Octonions", J. Algebra 220 (1999) 188–224). So a non-associative multiplication can indeed carry a cohomology class — a 3-*cocycle* valued in its associator, not a 2-cocycle valued in a unary cell. The correctly-typed question for **Norm** is therefore not about k at all but about the ternary associator Z_K of §4: *is Z_K (the coboundary of a suitable normalization cochain) a* 3*-cocycle on* **Norm***'s nerve?* We do not answer this here; we flag it as an **open** question and the natural degree-3 successor of this whole note.

The locus this reading probes is the *cochain* (single-morphism) level — and, as Remark 3.1 predicts, the invariant is not there.

3.2. **Reading 2: not a Karoubi splitting — a unary operation that preserves associativity.** Here the receptacle *exists*, which makes the negative subtler.

Lemma 3.4 (k lives in an associative hom-set and is idempotent)**.** *$n\colon DM \Rightarrow MD$ and $m\colon MD \Rightarrow DM$ are morphisms in the functor category $[\mathbf{Set},\mathbf{Set}]$, whose vertical composition is strictly associative. Hence $k = n\,;m \in \mathrm{End}_{[\mathbf{Set},\mathbf{Set}]}(DM)$, and "is k idempotent?" and "does k split?" are both well posed. Moreover k is idempotent: from the section identity $m\,;n = \mathrm{id}_{MD}$ (a normalized distribution, included as a subdistribution, already has unit mass, so renormalizing is the identity),*

$$k\,;k = (n\,;m)\,;(n\,;m) = n\,;(m\,;n)\,;m = n\,;\mathrm{id}_{MD}\,;m = n\,;m = k.$$

Theorem 3.5 (k is a split idempotent; registry `sesquilaw-magmoid-karoubi`, **proved**)**.** *$k = n\,;m$ is a* split *idempotent: it factors through MD via n and m with $m\,;n = \mathrm{id}_{MD}$ and $n\,;m = k$, the splitting object MD already present in $[\mathbf{Set},\mathbf{Set}]$. Consequently the Karoubi/Cauchy (idempotent) completion is a* no-op *on k — it already lies in the image of $[\mathbf{Set},\mathbf{Set}] \hookrightarrow \mathrm{Kar}([\mathbf{Set},\mathbf{Set}])$ — so there is no "missing splitting" and no splitting obstruction to carry the non-associativity.*

Proof. A split idempotent is a factorization $e = s\,;r$ through its image with $r\,;s = \mathrm{id}$; Lemma 3.4 gives exactly this through MD, with $m\,;n = \mathrm{id}_{MD}$ and $n\,;m = k$. The splitting object MD is already present, so idempotent completion adds nothing for k. □

What makes this decisive is the meta-lemma it instantiates.

Lemma 3.6 (idempotent completion preserves associativity)**.** *For any category $\mathcal{D}$, the Karoubi/Cauchy completion* $\mathrm{Kar}(\mathcal{D})$ *is a* category*: its composition is associative (Borceux I, §6.5; nLab, "idempotent completion").*

Corollary 3.7 (splitting can never measure non-associativity)**.** *Splitting idempotents in an associative $\mathcal{D}$ yields only an* associative *category. Hence no "split some idempotents in an associative core" construction can* produce *a non-associative composition; dually, the failure of a composition to associate can never be* identified with *the failure of an idempotent to split, because the latter is repaired by a construction (Lemma 3.6) that cannot repair the former.*

This is the "one reason" in its second reading: splitting an idempotent is a *unary* operation on a *single* morphism, and Corollary 3.7 says such an operation preserves associativity — so it never sees, let alone measures, the ternary defect. The home exists but carries no obstruction content; its locus is the *single idempotent morphism.*

Remark 3.8 (the vacuity is generic: Karoubi absorbs weakenings). An independent source, from a problem with no connection to **Norm**, confirms that this is a general feature of the weak-distributive-law family. Böhm–Lack–Street ("Idempotent Splittings, Colimit Completion, and Weak Aspects of the Theory of Monads", arXiv:1102.4931) show that weak monad notions whose defect is a *unit* failure — e.g. *demimonads*, which drop the unit law but keep associativity — reduce to *strict* structure inside the Cauchy-completed $Q_*\mathcal{K}$. That is: *Karoubi completion absorbs one-sided unit weakenings.* Their law (the unit) is not our law (associativity), so this does not transfer our verdict; but the shared moral is exactly Corollary 3.7's — Cauchy/Karoubi completion *absorbs* weakenings of the monad laws rather than *detecting* them, reached from a wholly different problem.

3.3. **Reading 3: not a lax distributive law — a fixed order cannot see a ternary incomparability.** The one technical (non-vocabulary) lead for a prior home of one-sidedness is *lax distributive laws by inequality* — Schoen, "Lax Liftings and Lax Distributive Laws" (arXiv:2307.16552), and the companion (T,V)-program of Tholen ("Lax Distributive Laws for Topology I/II", arXiv:1603.06251,

1605.04489). It is the subtlest negative: the home exists, is inhabited, and is genuinely one-sided — but one-sided *about the wrong thing*.

Definition 3.9 (lax distributive law, after Schoen)**.** Let P be a monad on **Set** and F an *endofunctor* (no algebraic structure). A *lax distributive law* is a transformation $\lambda_X \colon FP(X) \to PF(X)$ subject to three *one-directional inequalities* in the inclusion (support) order $\subseteq$ of relations (**Set** order-enriched to **Rel**, with $\mathbf{Rel}(X, Y) = P(X \times Y)$ a poset under $\subseteq$): lax naturality, and lax compatibility with the unit and multiplication of P. The laxified cells are the coherences of the *one* monad P; F is a bare functor and contributes none.

Claim 3.10 (registry: refuted; `handedness-not-lax-DL` **proved**). The non-associativity of **Norm** is not an instance of, and is not faithfully receivable by, a Schoen/Tholen lax distributive law.

We give the bookkeeping reason briefly and the conceptual kill in full.

Variance. Schoen's template needs (P monad, F bare functor, $\lambda \colon FP \to PF$); **Norm** supplies (D, M, m, n, k). Only two variance assignments type-check. With $P = M,\ F = D,\ \lambda = n \colon DM \to MD$, the map n carries the defect — but its failing coherence is D-multiplicativity, a coherence with the multiplication of $T = D$, which in Schoen's template (where $F = D$ is a *bare* functor) is *invisible*; the three laxified cells are all about $P = M$. With $P = D,\ F = M$, the only map of the right variance is the *strict* law m, carrying no defect. In neither assignment is the defect among the laxifiable cells. The same analysis applies to Tholen (same coherence-cell laxness, quantale-enriched).

The discriminator (two orders). The conceptual kill turns on which order one laxifies in, and is independent of the variance bookkeeping.

Lemma 3.11 (elementary)**.** *If $\mu \leq \nu$ pointwise for probability measures μ, ν on a set Y (total mass 1 each), then $\mu = \nu$.*

Proof. $\nu - \mu \geq 0$ pointwise and $\sum(\nu - \mu) = 1 - 1 = 0$; a nonnegative function of zero total is zero. □

Proposition 3.12 (the defect is a laxness in neither order)**.** *The associator defect of* **Norm** *is not a fixed-direction lax inequality — neither in the $\subseteq$ support order of Definition 3.9, nor in the pointwise sub-probability order.*

Proof. Support order. For a triple whose composite succeeds, both bracketings $(f \# g) \# h$ and $f \# (g \# h)$ are normalized distributions supported on the *same* set of reachable outputs (normalization rescales, it does not create or destroy support). So in the $\subseteq$ order the two bracketings are generically equal — the order under which Schoen/Tholen laxify simply *does not see* the defect; it is invisible.

Pointwise order. Pass to the finer pointwise order, where the defect *is* visible as distinct real vectors. A Schoen/Tholen laxness would require a *single fixed direction*, $(f \# g) \# h \leq f \# (g \# h)$ for all triples or the reverse for all. But both bracketings have total mass 1, so whenever they differ Lemma 3.11 forbids $\leq$-comparability — and non-associativity is precisely the regime where they differ. Neither fixed direction can hold: the defect lives strictly between $\leq$-incomparable elements.

Either order kills it: invisible in the one Schoen/Tholen use, $\leq$-incomparable in the one that sees it. □

Remark 3.13 (the honest near-miss, and why even it is not fixed-direction). There *is* a lax-looking cell in the DM pointwise order, $\mathrm{id}_{DM} \leq k$, since normalization raises mass on the successful outcomes ($k(\mu)(y) = \mu(y)/|\mu| \geq \mu(y)$ when $|\mu| \leq 1$). But the direction is not uniform across coordinates: at the failure/deficit coordinate $\perp$, normalization *removes* the failure mass, so there $k \leq \mathrm{id}$ — the inequality *reverses*. Even this one honest cell is not a single fixed-direction $\leq$. And in any case it is a statement about D's renormalization, not about $\#$ as a binary operation; a lax cell existing *somewhere* in the ambient order does not make the *associator defect* a laxness.

The locus is the *coherence* 2*-cell*: a fixed-direction order on one cell, blind to a ternary incomparability — the "one reason" in its third reading.

Remark 3.14 (honest scope of the Tholen half). Schoen arXiv:2307.16552 is deep-read; Tholen arXiv:1603.06251/1605.04489 is not directly read, and the no-match for Tholen rests on the *family-level* fact that its laxness is quantale-order laxness of coherence cells — the locus Proposition 3.12 rules out — not on any fine Tholen detail. Were a future read to surface a Tholen construction laxifying an *argument slot* of a composition rather than a coherence cell, this verdict should be revisited; we judge it unlikely but mark it the one unread dependency.

Movement II. The home it is

4. The positive home: an idempotent normalizer

If the defect is neither a cocycle, nor a splitting obstruction, nor a laxness, what *is* it? The honest answer is concrete and lives one level up from every map so far — at the binary operation itself. We isolate the minimal abstract structure that carries it. The structure turns out to be classical; we define it from scratch here and name its lineage in §9.

Fix a category $(\mathcal{A}, \cdot, 1)$ with strictly associative composition — think of the honest, associative ambient (substochastic kernels under matrix product; or the Kleisli category $\mathrm{Kl}(DM)$). Diagrammatic order: for $f \in \mathcal{A}(X,Y)$, $g \in \mathcal{A}(Y,Z)$, $f \cdot g \in \mathcal{A}(X,Z)$ is "f then g".

Definition 4.1 (normalizer, normalized magmoid)**.** A *normalizer* on $\mathcal{A}$ is a family of functions $K = K_{X,Y} \colon \mathcal{A}(X,Y) \to \mathcal{A}(X,Y)$ such that

- (K1) K is idempotent: $K(K(a)) = K(a)$;
- (U) K fixes identities: $K(1_X) = 1_X$;
- (LA) *left-absorption*: $K\big(K(x) \cdot c\big) = K(x \cdot c)$ for all composable x, c.

K is *two-sided* if additionally, for every $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} W$ with $f, g, h \in \operatorname{Im} K$,

- ($\mathsf{RA}_{\#}$) *right-absorption for* #: $K\big(f \cdot K(g \cdot h)\big) = K(f \cdot g \cdot h)$.

The *normalized magmoid* N_K has $\operatorname{Ob} N_K = \operatorname{Ob} \mathcal{A}$, $N_K(X,Y) := \operatorname{Im} K_{X,Y} = \{a : K(a) = a\}$, composition $f \# g := K(f \cdot g)$, and units $1^{\#}_X := 1_X$. We call N_K *handed* when it is non-associative.

Remark 4.2 (N_K is a magmoid, and ($\mathsf{RA}_{\#}$) is not the mirror of (LA)). $f \# g = K(f \cdot g) \in \operatorname{Im} K$ by (K1), and (U) gives units, so N_K is a magmoid; associativity is *not* assumed. Note that ($\mathsf{RA}_{\#}$) is *not* the formal mirror of (LA). (LA) is a *binary* law quantified over *all* x (its left argument ranges over $\mathcal{A}$, in particular over de-normalized products $f \cdot g$). ($\mathsf{RA}_{\#}$) is a *ternary* law quantified only over *normalized* $f, g, h \in \operatorname{Im} K$. They have different arities and different domains; the asymmetry between them is the whole phenomenon, not a notational accident.

Remark 4.3 (the laws are not vacuous). In (LA) the argument x ranges over products $x = f \cdot g$ of N_K-morphisms, which are generally *not* in $\operatorname{Im} K$ (composing two normalized kernels de-normalizes); where $K(x) \neq x$, (LA) is a genuine constraint. Likewise ($\mathsf{RA}_{\#}$) concerns the *un-normalized* product $y = g \cdot h$; it is *not* the triviality $K(f \cdot K(y)) = K(f \cdot y)$ for $y \in \operatorname{Im} K$.

5. The characterization, and the decisive closure criterion

Three results pin down N_K: a clean left bracketing, the equivalence "associativity = two-sided absorption", and — the statement that is directly checkable on any instance — a *closure criterion*. All use only $\cdot$ and K.

Theorem 5.1 (the left bracketing is clean)**.** *Let K be a normalizer. Then for all composable $f, g, h \in N_K$, $(f \# g) \# h = K(f \cdot g \cdot h)$.*

Proof. Apply (LA) with $x := f \cdot g$:

$$(f \# g) \# h = K\big(K(f \cdot g) \cdot h\big) \overset{\text{(LA)}}{=} K\big((f \cdot g) \cdot h\big) = K(f \cdot g \cdot h),$$

the last step by associativity of $\cdot$ in $\mathcal{A}$. The left bracketing normalizes once, at the very end. □

The proof uses only (LA) and associativity of $\cdot$. The right bracketing does not simplify this way, and the gap is the whole content.

Corollary 5.2 (associativity $\iff$ two-sided absorption)**.** *Let K be a normalizer. Then N_K is associative if and only if K is two-sided ((RA$_\#$) holds).*

Proof. For $f, g, h \in N_K$, Theorem 5.1 gives $(f\#g)\#h = K(f{\cdot}g{\cdot}h)$, while $f\#(g\#h) = K\big(f{\cdot}K(g{\cdot}h)\big)$ by definition. Associativity equates the two, which is exactly (RA$_\#$); conversely (RA$_\#$) equates them. □

(We present this as a corollary of Theorem 5.1: given the clean left bracketing it is pure unfolding. It was previously billed as a co-equal theorem, which overstated its depth.)

Corollary 5.3 (handedness is a genuine, non-vacuous obstruction)**.** *If K satisfies (LA) but not (RA$_\#$), then N_K is handed, and its associator defect equals exactly the defect $K(f \cdot g \cdot h)$ versus $K(f \cdot K(g \cdot h))$ of right-absorption — nothing more and nothing less.*

This is the sharp contrast with the Karoubi reading. There the receptacle was *vacuous*: k already splits, so idempotent completion measures nothing (§3.2). Here the receptacle is *inhabited*: Corollary 5.3 names a concrete, nonempty set of triples — those with $K(f \cdot K(g \cdot h)) \neq K(f \cdot g \cdot h)$ — that *is* the obstruction.

We now give the decisive, directly-checkable statement. Recall $\operatorname{Im} K$ is *closed* under $\cdot$ if $f, g \in \operatorname{Im} K$ composable $\Rightarrow f \cdot g \in \operatorname{Im} K$, i.e. $K(f \cdot g) = f \cdot g$.

Theorem 5.4 (closure criterion; registry `positive-home-one-sided-absorbing-idempotent`, **proved**)**.** *Let K be a normalizer. If $\operatorname{Im} K$ is closed under $\cdot$, then $f \# g = f \cdot g$ for all composable $f, g \in \operatorname{Im} K$, and N_K is associative. Equivalently, by contrapositive:*

$$\boxed{N_K \text{ handed} \implies \operatorname{Im} K \text{ is not closed under } \cdot}$$

Proof. If $\operatorname{Im} K$ is closed, then for composable $f, g \in \operatorname{Im} K$ we have $f \cdot g \in \operatorname{Im} K$, so $f \# g = K(f \cdot g) = f \cdot g$. Iterating, $(f \# g) \# h = (f \cdot g) \cdot h$ and $f \# (g \# h) = f \cdot (g \cdot h)$, equal by associativity of $\cdot$. The contrapositive is immediate. The proof uses only (K1) (to read $\operatorname{Im} K$ as the fixed-point set) and associativity of $\cdot$ — not even (LA). □

This is the decisive statement. The formalization it refutes is "handedness lives in a distinguished idempotent that left-absorbs, $K = \kappa \cdot (-)$": §7 shows such a K has closed image, so by Theorem 5.4 it can never be handed. But we must be equally clear about what the criterion does *not* say.

Theorem 5.5 (the converse fails — non-closure is necessary, not sufficient; **proved**)**.** *There exist normalizers K (satisfying (K1), (U), (LA)) with $\operatorname{Im} K$ not closed under $\cdot$ and yet N_K associative. Hence "$\operatorname{Im} K$ not closed $\Rightarrow$ handed" is false, and the full characterization of handedness remains (LA) $\wedge \neg$(RA$_\#$) (Corollary 5.2), of which non-closure is the clean* necessary *shadow.*

Witnesses (computed, exact). (a) *Minimal, order* 3. Let $M = \{e, a, b\}$ be the monoid with $a \cdot a = b$, $a \cdot b = b \cdot a = a$, $b \cdot b = b$ (associative, unit e; a non-invertible, so M is not a group). Put $K(e) = e$, $K(a) = a$, $K(b) = a$. Then (K1), (U), (LA) hold universally (checked over all 3×3 pairs), $\operatorname{Im} K = \{e, a\}$ is *not* closed ($a \cdot a = b \notin \operatorname{Im} K$), yet N_K on $\{e, a\}$ is the 2-element semilattice ($a \# a = K(b) = a$), which is associative. An exhaustive search shows no such example at order ≤ 2 and exactly six at order 3: order 3 is minimal.

(b) *Invertibility does not help.* Take $K = r$ the right-coset representative map of a subgroup $H \leq G$ with transversal T (so $\operatorname{Im} K = T$, $\# = \circ$). When $H \trianglelefteq G$ is normal, the quotient map restricts to a bijection $T \to G/H$ under which $\circ$ becomes the group law of G/H; so $(T, \circ) \cong G/H$ is a *group* (associative) for *every* transversal of a normal subgroup (such a T is itself a subgroup precisely when it is a *complement* of H). Explicitly for C_4, $H = \{0, 2\}$, $T = \{0, 1\}$: $\operatorname{Im} K$ is not closed ($1 + 1 = 2 \notin T$) yet $(T, \circ) \cong C_2$; likewise D_4 (12 transversals) and Q_8. □

So the honest picture is one-directional, with a strict gap:

$$\underbrace{\operatorname{Im} K \text{ closed}}_{\iff \# = \cdot|_{\operatorname{Im} K}} \overset{\text{strict}}{\Longrightarrow} \underbrace{(\mathsf{RA}_{\#}) \text{ holds}}_{\iff N_K \text{ associative}} .$$

The left "iff" (closure $\Leftrightarrow \# = \cdot|_{\operatorname{Im} K}$) is immediate from Theorem 5.4; Theorem 5.5 is the strictness of the middle arrow.

A word on how this sits against the weld's class $[\omega]$, to head off a tempting but wrong gloss. Over a group with $H \triangleleft G$, writing $s \cdot t = \varphi(s,t) \cdot (s \,\#\, t)$ with $\varphi(s,t) \in H$, there are *three* conditions, not two: closure of *this* transversal $\iff$ $\varphi \equiv 1$; *splitting* of $1 \to H \to G \to G/H \to 1$ $\iff$ $[\varphi] = 0 \in H^2(G/H; H)$ $\iff$ *some* transversal is closed; and associativity, which holds for *every* transversal because $(T, \#) \cong G/H$. So the split-vs-non-split axis is "*some* transversal closed vs. *none*" — entirely *inside* the associative regime — and is *not* the closure-vs-associativity axis. (The printed group witnesses of Theorem 5.5 take H the order-2 *centre* of C_4, Q_8, D_4 — a *non-split* central extension, 0 of whose transversals are subgroups — yet $(T, \#) \cong G/H$ is a group. For a general K on a monoid without inverses there is no φ at all.) This is *exactly* the reading of Remark 1.2: $[\omega]$ is a *splitting* obstruction of an *already-associative* weld, not an associativity obstruction. The closure criterion agrees with it; it does not relocate associativity into an H^2 class.

Remark 5.6 (the unified mechanism: closure and normality solve one criterion). Normality is *sufficient but not necessary* for $(T, \#)$ to associate: over $S_3 \times C_2$ there are 9 associative non-subgroup transversals with H non-normal; over S_4, 36. The sharp statement (companion note, registry `mirror-LA-sufficient-for-associativity`, **proved**) is a single criterion with two canonical solutions. For $f, g, h \in T$ write $g \cdot h = \varphi(g,h) \cdot (g \,\#\, h)$ with $\varphi(g,h) \in H$; then

$$(T, \#) \text{ associative} \quad \iff \quad f \cdot \varphi(g,h) \cdot f^{-1} \in H \quad \text{for all } f, g, h \in T$$

(0 mismatches over all 36 116 pairs (H, T) up to S_4). Its two canonical solutions are *closure* (T a subgroup, $\varphi \equiv 1$, criterion vacuous — Theorem 5.4) and *normality* ($H \triangleleft G$, so $f\varphi f^{-1} \in H$ automatically). Over a general base $\mathcal{A}$ the normality half is captured by the *mirror* of (LA), the law $K(c \cdot K(x)) = K(c \cdot x)$, which makes $\ker K$ a two-sided congruence and hence $N_K \cong \mathcal{A}/\ker K$; for cosets it holds $\iff$ $H \triangleleft G$. It is strictly stronger than $(\mathsf{RA}_{\#})$ and incomparable with closure — two genuinely different sufficient conditions, each strictly above associativity.

6. The instance: the normalized-kernel magmoid **Norm**

The abstract home is inhabited by the very object that forced the question.

Proposition 6.1 (**Norm** is a normalized magmoid)**.** *Let $\mathcal{A}$ be substochastic kernels: objects finite sets; morphisms nonnegative $X \times Y$ matrices with row sums ≤ 1; composition = matrix product. Let K be row-normalization, $K(M)_{xy} = M_{xy}/r_x$ with $r_x = \sum_{y'} M_{xy'}$, acting as the identity on zero-mass (failure/$\perp$) rows. Then $\operatorname{Im} K$ is the set of normalized kernels (each row a probability distribution of mass 1, or total failure $\perp$ of mass 0), and $N_K =$ **Norm** with $f \,\#\, g = K(f \cdot g)$ the Di Lavore–Román–Székely composition.*

Lemma 6.2 ((LA) holds for row-normalization)**.** *$K(K(x) \cdot c) = K(x \cdot c)$ for all composable substochastic x, c.*

Proof. On positive-mass rows $K(M) = D_M \cdot M$ with $D_M = \mathrm{diag}(1/r_i)$ a positive diagonal. *(1) Positive-diagonal invariance:* for any positive diagonal $P = \mathrm{diag}(p_i)$, $(P{\cdot}M)_{ij} = p_i M_{ij}$ has row sum $p_i r_i$, so $K(P{\cdot}M)_{ij} = p_i M_{ij}/(p_i r_i) = K(M)_{ij}$. *(2)* By associativity, $K(x){\cdot}c = (D_x{\cdot}x){\cdot}c = D_x{\cdot}(x{\cdot}c)$. Hence $K(K(x)\cdot c) = K(D_x \cdot (x \cdot c)) \overset{(1)}{=} K(x \cdot c)$. On zero-mass rows $K(x) = x$ and both sides agree. □

Note K is *not* $\kappa \cdot (-)$ for any fixed κ: the rescaling D_x depends on x. Non-principality does not by itself force $\mathrm{Im}\, K$ to be non-closed; but here $\mathrm{Im}\, K$ *is* non-closed (Lemma 6.3), which is exactly why (LA) is a genuine property and not an identity (contrast §7).

Lemma 6.3 ((RA$_\#$) fails; exact witness)**.** *There exist $f, g, h \in$ **Norm** with $K(f \cdot K(g \cdot h)) \neq K(f \cdot g \cdot h)$, so by Corollary 5.2* **Norm** *is handed.*

Proof. Over state sizes $1 \to 2 \to 3 \to 2$, take the normalized kernels

$$f = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \end{bmatrix}, \quad g = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}, \quad h = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \text{ (middle state } 1 = \bot).$$

Then $g \cdot h = \begin{bmatrix} \frac{2}{3} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$ has rows of *unequal* mass $\frac{2}{3}, \frac{1}{2}$ (the $\bot$ at the middle state destroys different amounts of mass on g's two rows). Computing,

$$(f \,\#\, g) \,\#\, h = K(f \cdot g \cdot h) = \begin{bmatrix} \frac{8}{11} & \frac{3}{11} \end{bmatrix} \text{ (Theorem 5.1)}, \qquad f \,\#\, (g \,\#\, h) = K\big(f \cdot K(g \cdot h)\big) = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \end{bmatrix},$$

which differ: (RA$_\#$) fails. Correspondingly $\mathrm{Im}\, K$ is not closed ($g \cdot h \notin \mathrm{Im}\, K$), as Theorem 5.4 requires. □

Remark 6.4 (why the middle must have dimension ≥ 3; minimal shape $1 \to 2 \to 3 \to 2$). The defect needs the inner normalization $K(g \cdot h) = D_{g \cdot h} \cdot (g \cdot h)$ to be a *non-scalar* diagonal — two middle rows rescaled by *different* factors — else $D_{g \cdot h} \propto I$ is absorbed by the outer K (Lemma 6.2, step (1)) and the defect vanishes. Non-scalar $D_{g \cdot h}$ requires the two middle rows of $g \cdot h$ to have unequal mass, which requires h to destroy unequal mass on them; with only a $\bot$ and one surviving outcome that forces a third middle state (one feeds $\bot$, two survive unequally). Hence the minimal shape is $1 \to 2 \to 3 \to 2$, exactly the witness above.

Remark 6.5 (recovery regime, after DLRS Prop. 4.9). The two bracketings agree when the inner normalization is trivial. Di Lavore–Román–Székely Proposition 4.9 (Sesquilaw magmoids) gives three sufficient conditions for $f \,\#\, (g \,\#\, h) = (f \,\#\, g) \,\#\, h$ in any sesquilaw (S, T, m, n): either (1) $f = u \,;\, S\eta_T$ for some u, (2) $g = v \,;\, S\eta_T$ for some v, or (3) $h = w \,;\, \eta_S T$ for some w. In the metalanguage gloss of their §5 these read, respectively, as f *partial*, g *partial*, or h *stochastic* (total). Our witness fails all three; the cleanest recovery is case (3), h total — then h loses no mass, so $g \cdot h$ and g have equal row masses, $D_{g \cdot h} = I$, and the inner normalization vanishes — and note the middle factor g need *not* be stochastic. A deterministic exact sweep (seed 20261019, 20000 trials) corroborates: 0 violations of (LA) and of Theorem 5.1; generic triples handed; and the predicted recovery regimes associative 5000/5000 — reproducing arXiv:2510.01131's Prop. 4.9 purity cases (each purity condition drives the defect to $\sim 2 \times 10^{-16}$, its negation to $\sim 10^{-2}$). These are **computed** corroborations; the **proved** content is Theorems 5.1,5.4,5.5, Corollary 5.2, and Lemmas 6.2–6.3.

Remark 6.6 (machine-checked, axiom-free). The construction is certified by a finite model (registry leaf `lean-positive-home-oracle`, **Lean-verified**). Row-normalization uses rational division (not kernel-decidable), so the **Norm** matrix instance stays computational (Python); to `decide` (LA) as a *universal* law and witness $\neg$(RA$_\#$) we formalise the *smallest* abstract monoid-with-normalizer realizing (LA) $\wedge \neg$(RA$_\#$) (none exists at order ≤ 3): $M = \{e, z, a, b\}$ with two-sided zero z, $a \cdot a = z$,

$a \cdot b = a$, $b \cdot b = b$, and $K = \mathrm{id}$ on $\{e, a, b\}$, $K(z) = b$. Lean `decide` checks: M is a monoid; (K1), (U); (LA) *universally*; Theorem 5.1; that ($\mathsf{RA}_{\#}$) *fails* and $\#$ is non-associative (witness a, a, a: $(a \# a) \# a = b \neq a = a \# (a \# a)$); Corollary 5.2 pointwise; the magmoid axioms; and the closure criterion on this model ($\mathrm{Im}\, K$ not closed). An `#print axioms` audit shows `[propext]` only — no `sorry`, no `Classical.choice`, no `native_decide`. This is the positive twin of the Karoubi-vacuity leaf `lean-split-idempotent-oracle`. (A pending Lean polish will print $b \cdot a = z$, add $b \cdot a = a$ as a breaking control, and record the order-4 uniqueness of the model — noted for a lean pass.)

7. The fourth map falls: a distinguished element cannot be handed

The classical template for a one-sided absorbing operation is a *near-semiring with a one-sided absorbing zero* 0 ($0 \cdot a = 0$ imposed, $a \cdot 0 = 0$ not). It suggests a fourth named map: formalize handedness as living in a distinguished *element*, i.e. take the normalizer to be left-multiplication by a fixed idempotent. Type it per family: a *principal* normalizer is $K_X(x) = \kappa_X \cdot x$ for idempotents $\kappa_X \in \mathcal{A}(X, X)$, $\kappa_X \cdot \kappa_X = \kappa_X$. The closure criterion dispatches it.

Proposition 7.1 (principal normalizers have closed image, hence are associative)**.** *A principal normalizer has* $\mathrm{Im}\, K$ *closed under* $\cdot$*; hence by Theorem 5.4* N_K *is associative. A distinguished idempotent element cannot carry handedness.*

Proof. For $a \in \mathrm{Im}\, K$ we have $\kappa \cdot a = a$ (if $a = \kappa x$ then $\kappa a = \kappa\kappa x = \kappa x = a$). So for composable $a, b \in \mathrm{Im}\, K$, $K(a \cdot b) = \kappa \cdot a \cdot b = (\kappa \cdot a) \cdot b = a \cdot b$, i.e. $a \cdot b \in \mathrm{Im}\, K$: $\mathrm{Im}\, K$ is closed. Theorem 5.4 gives associativity. □

Remark 7.2 (doubly dead)*.* Under the unit axiom (U) a principal normalizer is not merely associative but *trivial*: $K(1_X) = \kappa_X \cdot 1_X = \kappa_X$ must equal 1_X, so $\kappa_X = 1_X$ and $K = \mathrm{id}$. There is no nontrivial principal normalizer at all. So the element formalization dies twice over: it forces closed image (Proposition 7.1) and, with (U), forces the identity.

Proposition 7.1 is the fourth refutation, and it is the sharpest because it is not about a failed map into some theory but about the *kind of object* handedness can live on. An element indexes a *closed* image; handedness requires a *non-closed* one; these are incompatible. The row-normalization K of **Norm** is non-principal: its image is not closed (Lemma 6.3), which by Proposition 7.1 (contrapositive) already rules out principality.

8. One fact, four loci

The four refutations are not four coincidences. Each candidate lives at a *locus* at which the ternary associator is invisible:

named map	locus it probes	verdict	why (one fact, four reads)
$k \mapsto$ 2-cocycle	cochain / 1-cell	refuted	wrong arity: k is unary
$k \mapsto$ Karoubi splitting	idempotent morphism	refuted	unary; splitting preserves assoc.
$k \mapsto$ lax distributive law	coherence 2-cell	refuted	fixed order blind to $\leq$-incomp.
κ-element (near-semiring)	distinguished element	refuted	closed image $\Rightarrow$ assoc.
normalizer K	**argument slots of** $\#$	**receives it**	non-closed image; $\mathsf{LA} \wedge \neg(\mathsf{RA}_{\#})$

The first three candidates are all *unary* features of the single cell k — a coboundary, a splitting, an order on one morphism — and a unary feature cannot be the ternary associator. The fourth is a distinguished *element*, and an element indexes a *closed* image, which by the closure criterion forces associativity. What remains is the *argument slots of the binary operation* $\#$, where Definition 4.1 places K and Theorems 5.1–5.4 locate the defect. The mismatch is structural, not accidental: this is *why* the maps missed.

9. Provenance: the normalizer is the classical transversal construction

The surviving structure is *not* original ground, and we correct an earlier claim of ours that said it was. A dedicated literature sweep — initially mis-keyed on "near-semiring" and "absorptive morphism class", which returned a false null — resolves, once keyed on "transversal", "right loop", "right congruence", "exact factorization", into a classical construction.

Remark 9.1 (the normalizer = coset-representative map; # = the transversal right loop). For a subgroup $H \leq G$ with right transversal T ($1 \in T$, one representative per coset Hg), the representative map $r\colon G \to T$[1] is a retraction onto T satisfying *exactly* (K1), (U), (LA); the induced product $t \circ t' := r(t \cdot t')$ is our #, and $T = \operatorname{Im} r$. This is the classical construction of right loops from transversals: **Baer**, "Nets and groups", Trans. AMS 46 (1939) 110–141; **Lal**, "Transversals in groups", J. Algebra 181 (1996) 70–81; **Kiechle**, *Theory of K-Loops*, Springer LNM 1778 (2002). The oidified (categorical) version is already published: **Grabowski**, "An introduction to loopoids", arXiv:1510.03395 (Def. 4.1/Prop. 4.3: $t \bullet t' = \pi(t \cdot t')$; Ex. 1.1 writes the (LA) computation). Our closure criterion (Theorem 5.4, forward direction) is the classical "if the transversal is a subgroup then $\circ$ is the restriction of the group product"; cf. also arXiv:1309.7729, "On the Congruences in Right Loops". We emphasize, per Theorem 5.5, that the biconditional "the right loop is a group iff the transversal is a subgroup" is *false* in the presence of a nontrivial normal subgroup. This biconditional has *no source* — it was our own `state/PROVE.md` target, not a classical claim; the classical literature contains the forward direction *and* the quotient fact $H \triangleleft G \Rightarrow (T, \circ) \cong G/H$, but not the false iff.

Remark 9.2 (**Norm** $= D \cdot K$ is an exact factorization, *not* a Zappa–Szép product). The relation to the weld (§1) now sharpens, honestly. A normalized kernel factors (on its positive-mass rows) as $D \cdot K$ — a positive-diagonal rescaling D times a normalized K-image element — the probabilistic shadow of a group *exact factorization* $G = S \cdot H$ (every element a product of one factor from each). This is strictly *weaker* than a *Zappa–Szép* product, which requires *both* factors to be closed subgroups — and the whole point of this note is that the transversal factor $\operatorname{Im} K$ is *not* closed. So **Norm** sits in the same exact-factorization *family* as the weld, with the weld being the genuine Zappa–Szép case (both factors closed) and **Norm** one honest rung more general. That is all we claim: no structure map to the weld, and no relocation of the associativity defect into the weld's H^2 class (cf. Remark 1.3).

Remark 9.3 (honest novelty residue — narrow). What is *not* in the cited literature, and is ours, is narrow and we foreground only this:

(1) *A base without invertibility.* Baer, Lal, Kiechle and Grabowski all assume group/groupoid inverses. **Norm**'s substochastic kernels are non-invertible, and the order-3 witness of Theorem 5.5(a) is a non-invertible monoid. Running the transversal/right-loop construction over a general associative category/monoid *without* inverses is a modest extension of monoid right-congruence theory — a genuine but small gap in the cited sources, *not* a new phenomenon.

(2) *The instance.* The substochastic-kernel magmoid **Norm**, read as the exact factorization $D \cdot K$, is a legitimate novel *example* of the classical construction — not a novel construction.

A third candidate precedent, Grandis ("Categories with Negation", arXiv:1709.03186), whose left/right-absorbing morphism *classes* form an *ideal*, is structurally one level off (a class, not a

[1]The transversal/right-loop attributions of this remark (Baer; Lal; Kiechle; arXiv:1309.7729; Grabowski arXiv:1510.03395) were established by a dedicated literature search this cycle and are reported as the current state of evidence. The two arXiv entries are not yet logged at deep-read in the program's source index; a browse pass should record them before external submission — Grabowski's Example 1.1 in particular (the explicit (LA) computation) belongs on that deep-read list. The verdict "this construction is classical" does not depend on any single one of them.

single operator) and is not the home; we note it only to close the search. We would retract any novelty claim the moment a closer precedent surfaced.

10. Grant relevance

The program's obstruction-theory narrative gains a complete, honestly-bounded unit. For a concrete, finite-dimensional, machine-checkable witness of compositional non-associativity — the normalized substochastic magmoid **Norm**, an exact-factorization cousin of the central Zappa–Szép weld — we now know precisely *what the defect is not*: not a 2-cocycle, not a Karoubi-splitting obstruction, not a lax distributive law, not a distinguished near-semiring element; four named maps, decided, failing for one reason read four ways (a unary cell, or a closed-image element, cannot be a ternary associator). And we know *what it is*: a one-sided absorptivity received by an idempotent normalizer K, with "associativity = K two-sided absorbing" (Corollary 5.2) a theorem, with the decisive, directly-checkable *closure criterion* "handed $\Rightarrow$ $\operatorname{Im} K$ not closed" (Theorem 5.4), with **Norm** as a faithful instance, and with the whole package identified — honestly — as the classical transversal construction of right loops, placing the invariant in the program's exact-factorization theory (of which the Zappa–Szép weld is the two-sided-closed case). The one open successor, flagged as such, is the correctly-typed cohomological question of Remark 3.3: is the ternary associator Z_K a degree-3 class? "Normalization's non-associativity is a handedness datum of a transversal normalizer, not a characteristic class of k" is a crisp, defensible statement about where compositional non-associativity lives — and the kind of negative-plus-positive result, with provenance stated rather than overclaimed, that keeps the Impact narrative honest.

Kodamai